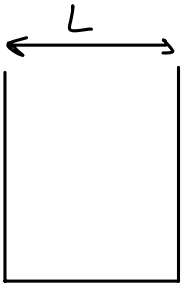
④



$$E_n = \frac{n^2 h^2}{8mL^2}$$

a, b) $$Z_1 = \sum_i e^{-\beta E_i} = \sum_n e^{-\beta \frac{n^2 h^2}{8mL^2}} \approx$$

$$\approx \int_0^\infty e^{-\frac{\beta h^2 x^2}{8mL^2}} dx = \int_0^\infty e^{-y^2} dy \sqrt{\frac{8mL^2}{\beta h^2}} =$$

$$y = \sqrt{\frac{\beta h^2}{8mL^2}} \cdot x \qquad \int_0^\infty e^{-y^2} dy = \frac{\sqrt{\pi}}{2}$$

$$= \sqrt{\frac{2\pi m L^2}{\beta h^2}} = \sqrt{\frac{2\pi m k_B T}{h^2}} \cdot L$$

c) three directions: $$Z_1^{(3D)} = V \cdot \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}$$

indistinguishable particles: $$Z = \frac{(Z_1^{(3D)})^N}{N!} = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2}\right)^{3N/2}$$

⑤

a) $$p = \frac{2\Delta N m \langle v_x \rangle}{A \Delta t} = 17500\ \text{Pa} \qquad \left(m = \frac{M}{N_A}\right)$$

b) $$\frac{1}{2} m \langle v_x^2 \rangle = \frac{1}{2} k_B T \Rightarrow T = \frac{m \langle v_x^2 \rangle}{k_B} \approx 300\,\text{K}$$

Two atomic molecule: $\varepsilon = \frac{5}{2} k_B T \approx 10^{-20}$ J

c) $pV = N k_B T \Rightarrow \frac{V}{N} = \frac{k_B T}{p}$ → volume of a cube around a particle



average distance: $$\lambda = \sqrt[3]{\frac{k_B T}{p}} \approx 6{,}2\ \text{nm}$$

⑥ a) Air: mainly $O_2 + N_2$ ⇒ average kinetic energy of the particles: $\varepsilon = \frac{5}{2} k_B T$

$$\Delta \varepsilon = \frac{5}{2} k_B \Delta T = 5{,}2 \cdot 10^{-22}\ \text{J}$$

b) $$U = N\varepsilon = \frac{5}{2} N k_B T = \frac{5}{2} pV = \text{const}$$

(7)

$$\frac{1}{2}mv_{rms}^2 = \frac{1}{2}m\langle v_x^2 + v_y^2 + v_z^2\rangle = \frac{1}{2}m\langle v_x^2\rangle + \frac{1}{2}m\langle v_y^2\rangle + \frac{1}{2}m\langle v_z^2\rangle = \frac{3}{2}k_BT$$

$$v_{rms} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3k_BT}{M/N_A}} = \sqrt{\frac{3RT}{M}}$$

$$\Delta T = \frac{M}{3R}\cdot\left(v_2^2 - v_1^2\right) \Rightarrow M = \frac{3R\Delta T}{v_2^2 - v_1^2} = 28\,\frac{g}{mol} \Rightarrow N_2 \text{ for example}$$